

Topic B3

Movement into and out of cells

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B3.1	Diffusion
B3.2	Osmosis
B3.3	Active transport

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Diffusion

Diffusion is the net movement of particles from higher to lower concentration due to random motion. A steeper concentration gradient, higher temperature and larger surface area increase rate; a shorter distance also increases exchange.

Examiner insight

Describe net movement, because particles move randomly in both directions.

2. Osmosis

Osmosis is the net movement of water molecules from higher water potential to lower water potential through a partially permeable membrane. Dilute solution has higher water potential than concentrated solution.

Examiner insight

Do not define osmosis using only “high concentration to low concentration”; specify water and water potential.

3. Effects on cells

Plant cells in dilute solution gain water and become turgid because the wall resists expansion. In concentrated solution they lose water and may plasmolyse. Animal cells can burst in very dilute solution or crenate in concentrated solution because they lack a cell wall.

Examiner insight

Turgor supports non-woody plants; the cell wall prevents bursting.

4. Active transport

Active transport moves substances against a concentration gradient using energy from respiration and carrier proteins in membranes. Root hair cells absorb mineral ions even when soil concentration is lower than cell concentration; intestinal cells can absorb glucose against its gradient.

Examiner insight

Evidence for active transport includes movement against a gradient and inhibition when respiration is reduced.

Worked application

Situation: Root cells contain more nitrate ions than surrounding soil.

Prediction: Nitrate uptake occurs by active transport.

Reason: Ions move against the gradient using respiratory energy and membrane carriers.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B3

ovement into and out of cells

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement best defines diffusion? [1]
- A net movement from higher to lower concentration due to random motion
 - B net movement of water from higher to lower water potential through a partially permeable membrane
 - C movement against a concentration gradient using energy
 - D a plant cell firm from water uptake and pressure against its wall
2. Which statement best defines osmosis? [1]
- A movement against a concentration gradient using energy
 - B net movement of water from higher to lower water potential through a partially permeable membrane
 - C a plant cell firm from water uptake and pressure against its wall
 - D the cell membrane pulling away from the wall after water loss
3. Which statement best defines active transport? [1]
- A a plant cell firm from water uptake and pressure against its wall
 - B the cell membrane pulling away from the wall after water loss
 - C movement against a concentration gradient using energy
 - D net movement from higher to lower concentration due to random motion
4. Which statement best defines turgid? [1]
- A the cell membrane pulling away from the wall after water loss
 - B net movement from higher to lower concentration due to random motion
 - C net movement of water from higher to lower water potential through a partially permeable membrane
 - D a plant cell firm from water uptake and pressure against its wall
5. Which statement best defines plasmolysis? [1]
- A the cell membrane pulling away from the wall after water loss
 - B net movement from higher to lower concentration due to random motion
 - C net movement of water from higher to lower water potential through a partially permeable membrane
 - D movement against a concentration gradient using energy
6. Which statement about diffusion is correct? [1]
- A Movement requiring energy against a gradient.
 - B Diffusion is the net movement of particles from higher to lower concentration due to random motion.
 - C Movement of any solute through a cell wall.
 - D Passive movement down a gradient.

7. Which statement about osmosis is correct? [1]
- A Movement of any solute through a cell wall.
 - B Passive movement down a gradient.
 - C Osmosis is the net movement of water molecules from higher water potential to lower water potential through a partially permeable membrane.
 - D A cell whose membrane has pulled from the wall.
8. Which statement about effects on cells is correct? [1]
- A Passive movement down a gradient.
 - B A cell whose membrane has pulled from the wall.
 - C Bursting of an animal cell.
 - D Plant cells in dilute solution gain water and become turgid because the wall resists expansion.
9. Which statement about active transport is correct? [1]
- A Active transport moves substances against a concentration gradient using energy from respiration and carrier proteins in membranes.
 - B A cell whose membrane has pulled from the wall.
 - C Bursting of an animal cell.
 - D Movement requiring energy against a gradient.
10. Which prediction is most appropriate when root cells contain more nitrate ions than surrounding soil? [1]
- A Movement requiring energy against a gradient.
 - B Passive movement down a gradient.
 - C Nitrate uptake occurs by active transport.
 - D Bursting of an animal cell.

Section B · Structured questions

11 Biologists use precise vocabulary when studying movement into and out of cells.

(a) Define diffusion.

[2]

(b) Define osmosis.

[2]

(c) Distinguish active transport from the incorrect idea: “passive movement down a gradient”.

[2]

(d) State two important features of diffusion.

[2]

[Total: 8]

12 Root cells contain more nitrate ions than surrounding soil.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records percentage mass change under three conditions.

condition	percentage mass change
0.0 mol/dm ³	18
0.4 mol/dm ³	1
0.8 mol/dm ³	-16

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to place equal potato cylinders in sucrose solutions of different concentration.

(a) Identify the independent variable.

[1]

(b) Identify the dependent variable.

[1]

(c) State two control variables.

[2]

(d) Describe how to obtain reliable results across a suitable range.

[3]

(e) Suggest one specific improvement.

[1]

[Total: 8]

15 Apply your knowledge of movement into and out of cells to an unfamiliar context.

(a) Define turgid.

[2]

(b) Explain the biological importance of effects on cells.

[3]

(c) Correct this statement: “bursting of an animal cell”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. Net movement from higher to lower concentration due to random motion.
- 2 B. Net movement of water from higher to lower water potential through a partially permeable membrane.
- 3 C. Movement against a concentration gradient using energy.
- 4 D. A plant cell firm from water uptake and pressure against its wall.
- 5 A. The cell membrane pulling away from the wall after water loss.
- 6 B. Describe net movement, because particles move randomly in both directions.
- 7 C. Do not define osmosis using only “high concentration to low concentration”; specify water and water potential.
- 8 D. Turgor supports non-woody plants; the cell wall prevents bursting.
- 9 A. Evidence for active transport includes movement against a gradient and inhibition when respiration is reduced.
- 10 C. Ions move against the gradient using respiratory energy and membrane carriers.

Solutions · Structured questions

11. (a) net movement from higher to lower concentration due to random motion [2]. (b) net movement of water from higher to lower water potential through a partially permeable membrane [2]. (c) movement against a concentration gradient using energy; not passive movement down a gradient [2]. (d) Credit any two accurate features from the guide: Diffusion is the net movement of particles from higher to lower concentration due to random motion. A steeper concentration gradient, higher temperature and larger surface area increase rate; a shorter distance also increases exchange. [2].

12. (a) Nitrate uptake occurs by active transport [2]. (b) Ions move against the gradient using respiratory energy and membrane carriers [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $18 - -16 = 34$ [2]. (b) The values change from 18 to -16, with the highest value 18. [2]. (c) Increasing external solute concentration lowers water potential and changes net water movement from gain to loss [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Sucrose concentration [1]. (b) Percentage change in mass [1]. (c) Same cylinder size, time, temperature and potato source [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Blot cylinders consistently before weighing and repeat [1].

15. (a) a plant cell firm from water uptake and pressure against its wall [2]. (b) Award linked points from: Plant cells in dilute solution gain water and become turgid because the wall resists expansion. In concentrated solution they lose water and may plasmolyse. Animal cells can burst in very dilute solution or crenate in concentrated solution because they lack a cell wall. [3]. (c) Correct statement: plasmolysis means the cell membrane pulling away from the wall after water loss [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.